



Lab 1: Introduction to Networking and Cisco Packet Tracer

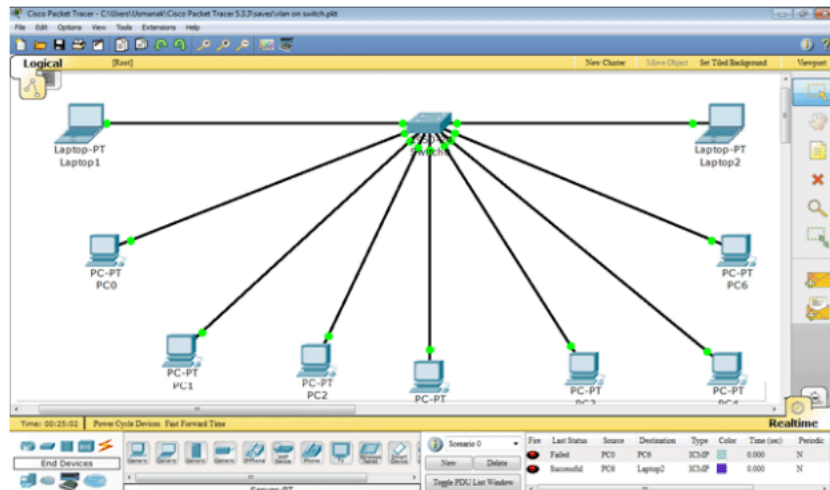
Introduction to Networking

A **computer network** is a collection of interconnected devices that communicate with each other to share resources and data. Networks play a vital role in modern computing by enabling collaboration, communication, and resource sharing, such as accessing files, printers, and internet services.

Networks can range from small **Local Area Networks (LANs)**, which connect devices within a single building, to vast **Wide Area Networks (WANs)** that span across cities, countries, or even continents. The primary purpose of these networks is to ensure communication between devices and optimize resource usage.

As networks grow in complexity, learning how to design, configure, and troubleshoot them is essential. To achieve this, network simulation tools like **Cisco Packet Tracer** are invaluable for building and understanding networks in a controlled, virtual environment.

Introduction to Cisco Packet Tracer



Cisco Packet Tracer's interface.

Cisco Packet Tracer is a powerful, interactive network simulation tool developed by Cisco Systems. It provides a hands-on platform for designing network topologies, configuring devices, and troubleshooting issues, all without needing physical hardware.

Whether you're a beginner or an advanced learner, Packet Tracer simplifies complex networking concepts by visualizing how devices interact and communicate. Its user-friendly interface allows



you to experiment freely and test configurations in a risk-free environment, making it an essential tool for networking students and professionals.

In this lab, you'll learn how to use Cisco Packet Tracer to create and manage a basic network topology, helping you understand foundational networking principles while gaining practical skills with this versatile tool.

Exploring the Cisco Packet Tracer Interface

When you open Cisco Packet Tracer, you will see a user-friendly interface with various menus, tools, and workspaces. Here's a brief overview of the key components:

1. **Menus and Toolbars:**
 - a. The top menu bar provides access to file operations, device selection, and simulation controls.
 - b. The toolbar contains shortcuts to commonly used tools, such as selecting devices, connecting cables, and deleting objects.
2. **Workspaces:**
 - a. **Logical Workspace:** This is where you design and simulate your network topology. You can drag and drop devices, connect them, and configure their settings.
 - b. **Physical Workspace:** This allows you to visualize the physical layout of your network, such as where devices are placed in a building or data center.
3. **Simulation Mode:**
 - a. Packet Tracer offers a simulation mode where you can observe how data packets travel across the network. This is useful for understanding how devices communicate and troubleshoot issues.

Creating a Simple Network Topology

A network topology refers to the arrangement of devices and connections in a network. In this lab, you will create a simple topology consisting of two PCs connected to a switch.

Steps:

- Drag and drop two PCs and one switch from the device menu onto the logical workspace.
- Use straight-through cables to connect the PCs to the switch. These cables are typically used to connect different types of devices, such as a PC to a switch (2960).
- Assign IPv4 addresses to each PC manually. For example:
 - PC1: 192.168.1.1
 - PC2: 192.168.1.2
- Ensure that both PCs are on the same subnet (e.g., subnet mask 255.255.255.0).



Verifying Connectivity

Once the network is set up, you need to verify that the devices can communicate with each other.

Using the ping Command:

- Open the command prompt on one PC and use the ping command to test connectivity with the other PC. For example: ping 192.168.1.2.
- A successful ping indicates that the devices are correctly connected and can communicate.

Common Errors/Issues:

- **Wrong cable type:** Ensure you are using the correct cable (e.g., straight-through for PC-to-switch connections).
- **IP address mismatch:** Verify that the IP addresses are correctly configured and on the same subnet.

Documenting the Network

Proper documentation is essential for understanding, maintaining, and troubleshooting a network. In this lab, you will document your network topology using labels and notes.

To annotating the topology you made, you can

1. Use labels to identify each device (e.g., PC1, PC2, Switch1), for this go to Options > Preferences > Tick the boxes for Show Device Model Labels and Device Name Labels.
2. Add notes to specify the IP addresses assigned to each device.

Lab Tasks

CLO1: Explain the fundamental principles, architectures, and communication models of computer networks, highlighting the roles of different layers in end-to-end data exchange.

Note: All tasks must be documented in a single report.

Task 1: Explore the Cisco Packet Tracer Interface (5 Marks)

Explore the Cisco Packet Tracer interface by opening the software and navigating through its menus, tools, and workspaces to become familiar with its layout and functionality. Focus on identifying key features such as the **Simulation Mode**, which allows you to visualize network packet flow, and the **Logical Workspace**, where you design and configure network topologies.



Additionally, examine the **Physical Workspace**, which provides a real-world representation of network setups. Then, take a clear screenshot of the interface and label the following key components: Menu Bar, Tool Bar, Device Type Selection Box, Workspaces (Logical and Physical), Add Simple PDU (P) and Place Note (N). Include the screenshot in your report.

Task 2 Create a Simple Network Topology and verify connectivity (5 Marks)

Create a network topology in Cisco Packet Tracer by adding two PCs and one switch (2960-24TT) to the workspace. Connect each PC to the switch using appropriate cables, such as straight-through cables for different devices, to establish physical connectivity.

Manually configure IPv4 addresses for the devices; assign **192.167.1.5** to PC1 and **192.167.1.6** to PC2, ensuring they are within the same subnet. Use the **ping** command from PC1 to test connectivity with PC2 and verify successful communication. Document your steps in the report, clearly showing how you configured the PCs and carried out the ping command test.

Next, introduce a common issue, such as using a crossover cable instead of a straight-through cable or assigning mismatched IP addresses that fall outside the same subnet. Document the troubleshooting process, including identifying the issue, testing configurations, and correcting it by replacing the cable or reconfiguring the IP addresses, and then retesting connectivity to confirm the resolution (2-3 lines are more than enough).

Task 3: Documenting the Network (5 Marks)

Annotate the network topology created in Tasks 2 and 3 by adding labels and notes to identify each device and its corresponding IP address. Clearly label the devices as **PC1**, **PC2**, and **Switch**, and indicate the IP addresses (e.g., **192.168.1.1** and **192.168.1.2**) and cable types used. Take a screenshot of the fully annotated topology and include it in the report.

Also write a concise description of the network, explaining its purpose, such as demonstrating basic connectivity and troubleshooting. Include a summary of each device's role, the type of connections used, and how the setup enables communication between the devices (a few lines are more than enough).

Submission Guidelines (5 Marks):

- Submit a single PDF document containing all tasks and the CISCO packet files to the GCR, **ensure your file name is visible in all pictures and contains your roll number**.
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **L_i231234_Lab1**.
- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **L_i231234_Lab1**.



CL3001 - Computer Networks Lab

Instructor: Ms. Zoya Mahboob

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- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.
- In case a demo is conducted, there will be deductions based on your answers to the questions asked.

NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.

Sample screenshot:

